

Dani Roytburg

CURRICULUM VITAE

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Master's student in the Machine Learning Department at Carnegie Mellon University focused on reliable and interpretable AI, multi-agent training paradigms, and activation-based model analysis.

Education

Carnegie Mellon University, School of Computer Science

expected May 2027

M.S. in Machine Learning

Emory University, College of Arts and Science

2025

B.S. in Computer Science; Quantitative and Social Sciences · *summa cum laude*

Research & Professional Experience

ML Alignment and Theory Scholars

May 2026 – September 2026

Research Fellow · Berkeley, CA

Advised by Shi Feng. Developed persona-training methods which emulate midtraining for better model organisms and constitutions. Engineered evaluations across 100+ training runs on synthetic-document pipelines. Evaluated how tastefully frontier models can test ideas in safety research.

Carnegie Mellon University

September 2025 – Present

Research Assistant, DI Group · Pittsburgh, PA

Develop multi-agent reinforcement learning environments to improve legibility of reasoning traces. Design user studies and adversarial setups to measure influence of legibility on human and multi-agent interactions.

Martian Research

May 2025 – February 2026

Research Fellow · San Francisco, CA

Engineered activation profiles of language model evaluators to suppress self-preferential bias, applied to LLM routers (ICML 2026, MechInterp @ NeurIPS 2025). Winner of Mechanistic Router Interpretability Hackathon (40 submissions).

Emory University

November 2021 – May 2025

Research Assistant, Digital Humanities Lab · Atlanta, GA

Designed and trained large-scale graph-text models; evaluated adversarial robustness of LLMs; built historical correspondence networks and dashboards.

ZS Associates

June 2024 – August 2024

Internship - Business Technology Solutions · Evanston, IL

Automated ETL pipelines and implemented synthetic-data imputation for compromised client data.

Cloverpop

January 2024 – May 2024

Intern — Machine Learning · Chicago, IL

Deployed decision-management systems and structured-prediction models on meeting transcripts.

McCormick and Company

June 2023 – August 2023

Intern — Global Marketing and Sales Data Science · Hunt Valley, MD

Built Amazon sales data visualizations and automated analytics with GCP BigQuery and BERT-based embedding search.

Northeastern University

June 2022 – October 2022

Research Assistant, Mediacloud Lab · Boston, MA

Extended NER to long-tail ethnics cuisines using Bon Appetit data.

Publications

* Equal contribution / co-first authorship.

Under review

Pre-training interventions, ex post facto: Grafting model beliefs across checkpoints

Peter Nutter*, **Dani Roytburg***, Clément Dumas, Jinghua Ou, Shi Feng.

Under review, 2026.

[Paper](#) · [Blog post](#) · [Project website](#) · [Code](#)

Conference & workshop papers

Are LLM Evaluators Really Narcissists? Sanity Checking Self-Preference Evaluations

Dani Roytburg, Matthew Bozoukov, Matthew Nguyen, Jou Barzdukas, Mackenzie Puig-Hall, Narmeen Oozeer.

Forty-Third International Conference on Machine Learning, 2026.

[Paper](#) · [Code](#)

Measuring Weak-to-Strong Legibility of Reasoning Models

Dani Roytburg, Shreya Sridhar, Daphne Ippolito.

Advances in Neural Information Processing Systems (NeurIPS), 2026.

Also at: ICML 2026 Workshop on AI for Good (AI4GOOD).

[Paper](#)

Disentangling Models from Personas in Heterogeneous LLM Simulations

Dani Roytburg, Daphne Ippolito.

Second Workshop on Social Simulation with LLMs: Fidelity in Applications (COLM 2026, Spotlight), 2026.

[Paper](#) · [Code](#)

Mind the Gap: Pathways Towards Unifying AI Safety and Ethics Research

Dani Roytburg*, Beck Miller*.

Proceedings of the International Association for Safe and Ethical AI, 2026.

[Paper](#) · [Code](#)

Breaking the Mirror: Activation-Based Mitigation of Self-Preference in LLM Evaluators

Dani Roytburg*, Matthew Bozoukov*, Hongyu Fu, Matthew Nguyen*, Jou Barzdukas*, Narmeen Fatimah Oozeer.

Mechanistic Interpretability Workshop at NeurIPS 2025, 2025.

Also at: NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle; NeurIPS 2025 Workshop on Reliable ML from Unreliable Data.

[Paper](#) · [Code](#)

Words and Action: Modeling Linguistic Leadership in #BlackLivesMatter Communities

Dani Roytburg*, Deborah Olorunisola*, Sandeep Soni, Lauren Klein.

Proceedings of the International AAAI Conference on Web and Social Media, 2025.

[Paper](#)

Awards

Best Poster, LTI Student Research Symposium

2026

Emory University Dean's List

2022, 2024, 2025

Winner, Martian Research Mechanistic Interpretability Hackathon

2025

Technical Skills

Python · Java · R · JavaScript · Typescript · SQL · PyTorch · JAX · HuggingFace Transformers · scikit-learn · spaCy · networkx · D3.js · React.js · GCP · Docker · MySQL